Class 12 Pt2.

Parisa Charoensook A17939591

> Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

The sample sizes for each genotype are A/A = 108, A/G = 233, and G/G = 121. The median expression levels were A/A = 31.25, A/G = 25.06, G/G = 20.07.

```
geno <- read.table("expression.txt" ,
header = TRUE)
```

```
table(geno$geno)
```

```
A/A A/G G/G
108 233 121
```

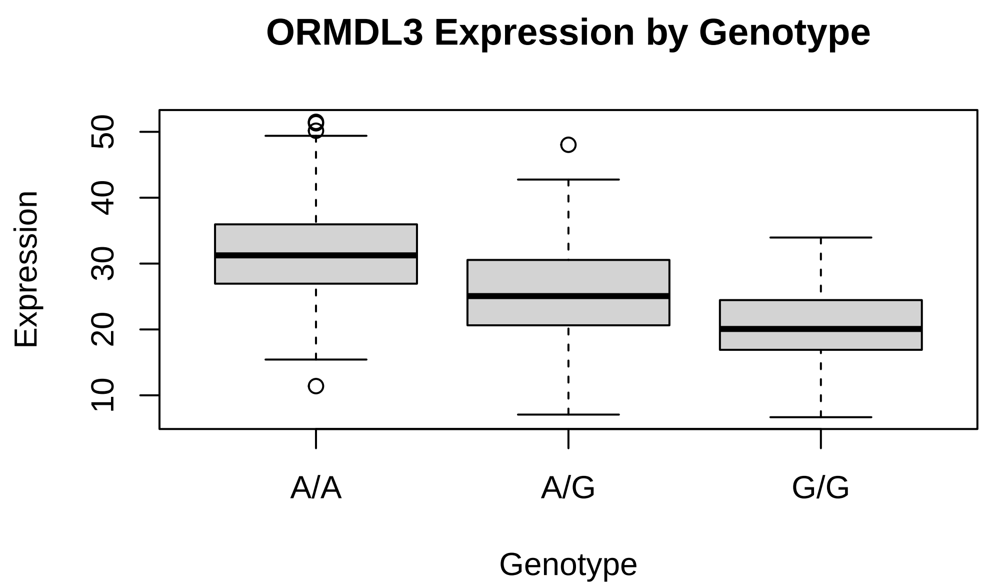
```
tapply(geno$exp, geno$geno, median)
```

```
     A/A      A/G      G/G
31.24847 25.06486 20.07363
```

> Q14. Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

The boxplot shows that ORMDL3 expression decreases from A/A to A/G to G/G. Individuals with A/A genotype have highes expressions and G/G has least. This sugggests that the SNP is associated with changes in ORMDL3 expression and has an affect on gene expression.

```
boxplot(exp ~ geno, data=geno,
        main = "ORMDL3 Expression by Genotype",
        xlab = "Genotype",
        ylab = "Expression")
```


ORMDL3 Expression by Genotype
Expression
10
20
30
40
50
A/A
A/G
G/G
Genotype